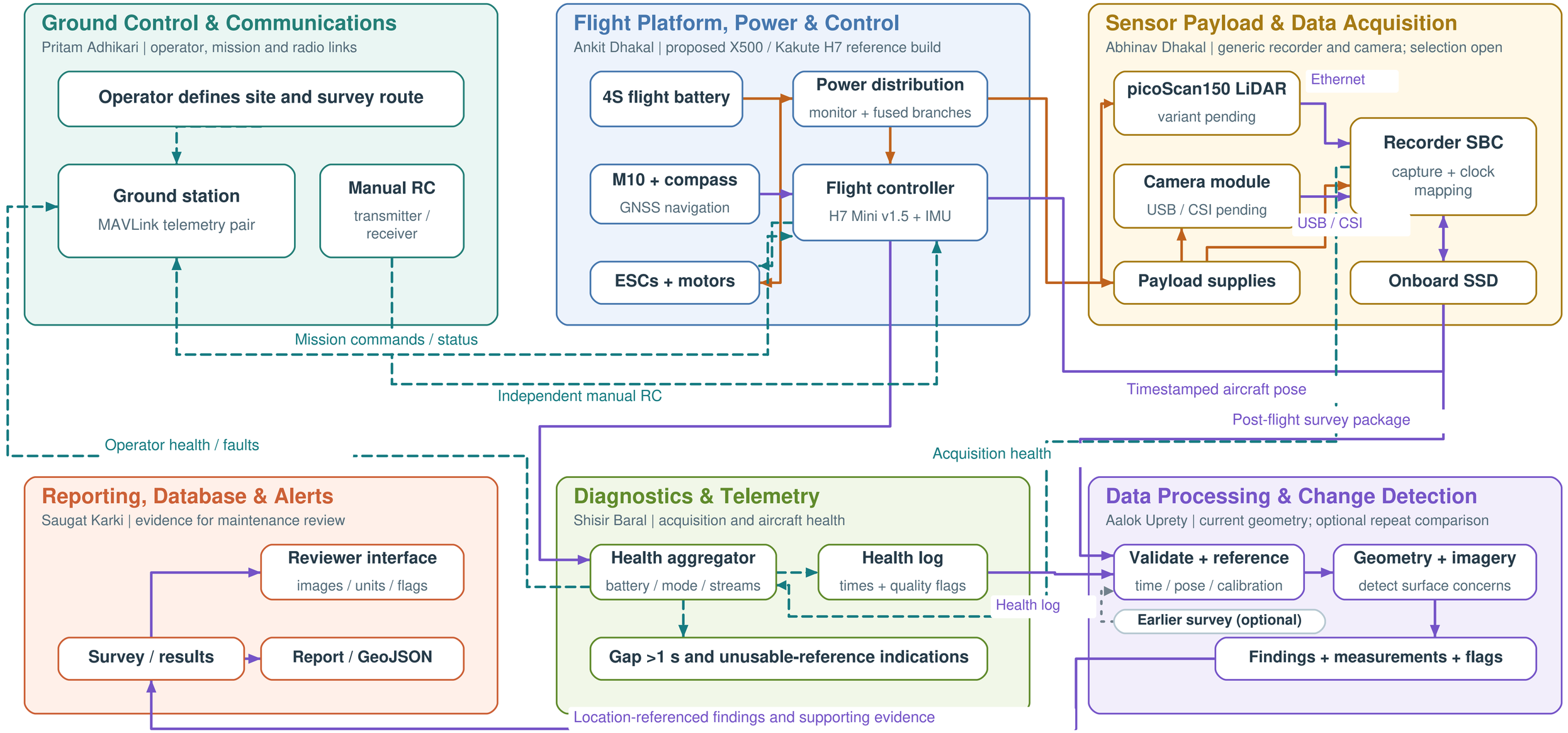
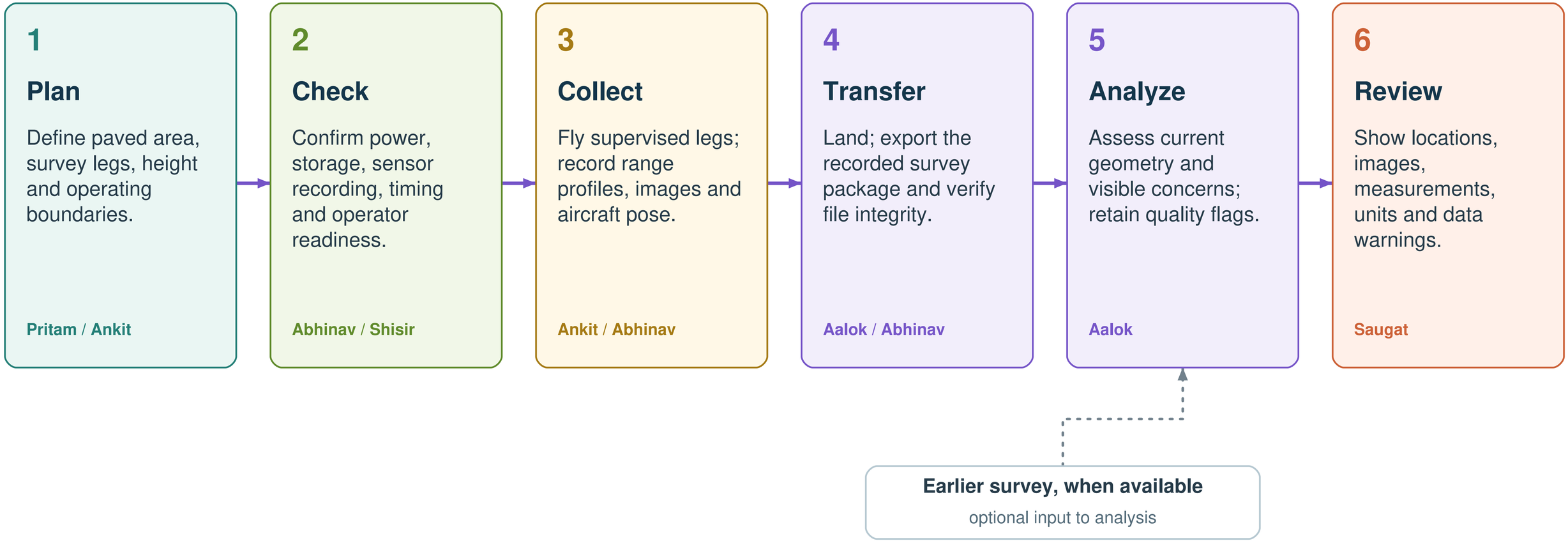
PaveWatch | System architecture

Six subsystems, major components, and the connections needed for a referenced pavement survey.



PaveWatch | From a survey to inspectable evidence

A process overview to introduce the system before discussing components.



Baseline deliverable: one current survey with evidence. Historical comparison adds context when available.

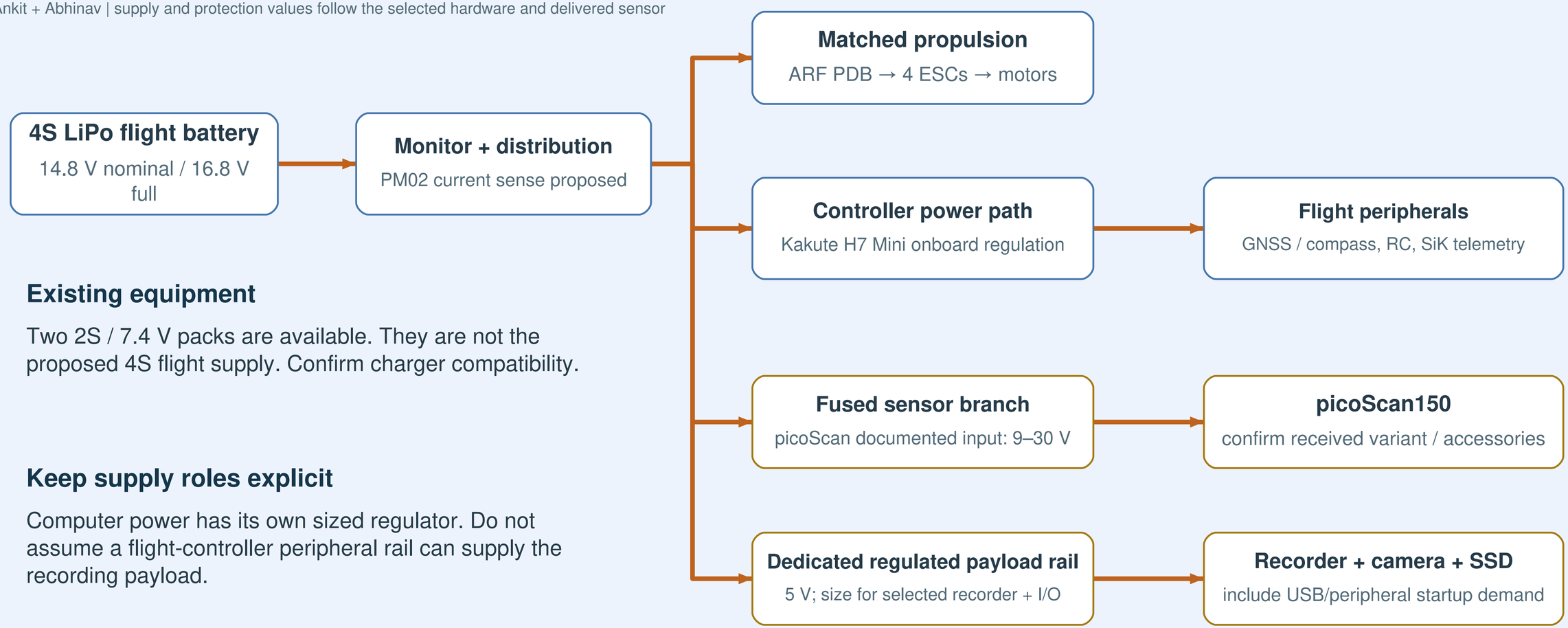
Proposed envelope: 5–8 m above pavement, 1–2 m/s, dry daylight • Performance limits require validation.

PaveWatch | Power architecture

Proposed 4S flight build with separately protected payload supplies; conceptual power paths, not a wiring schematic.

Flight and payload power domains

Ankit + Abhinav | supply and protection values follow the selected hardware and delivered sensor



Existing equipment

Two 2S / 7.4 V packs are available. They are not the proposed 4S flight supply. Confirm charger compatibility.

Keep supply roles explicit

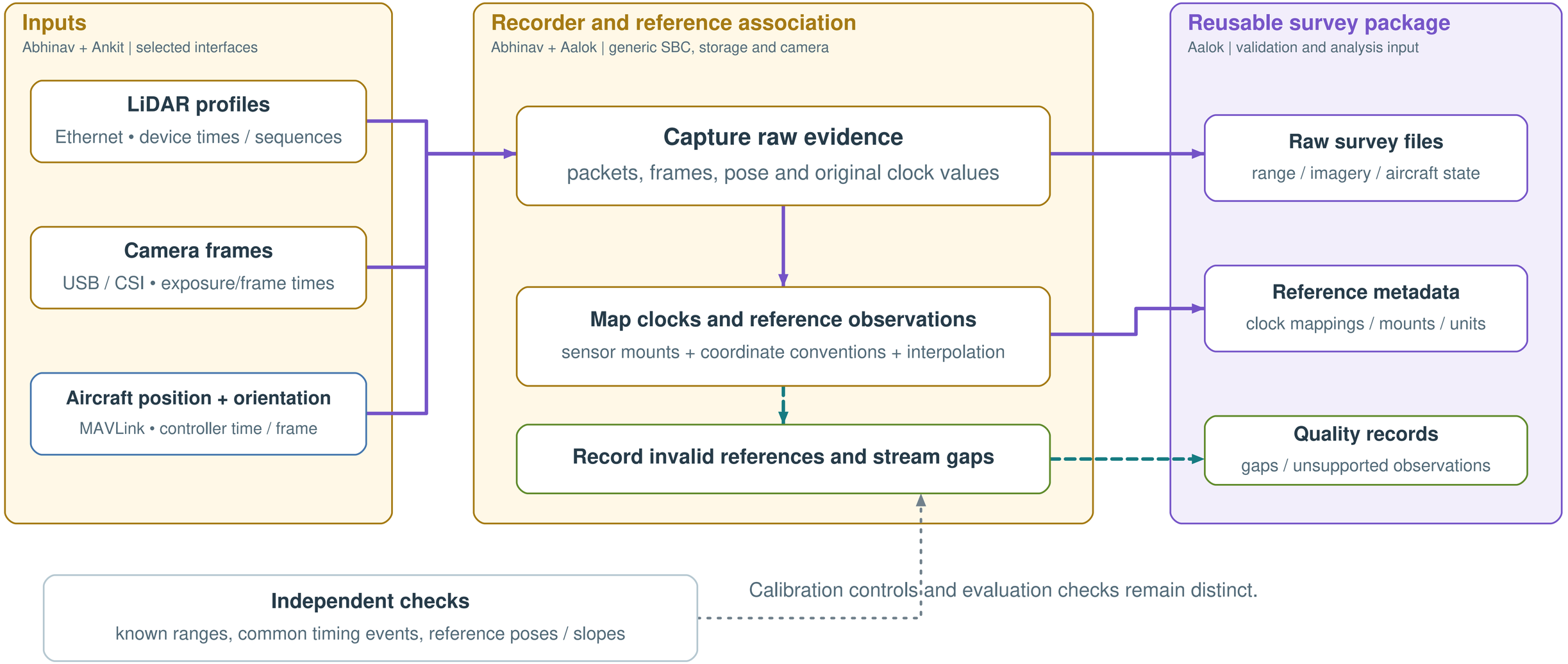
Computer power has its own sized regulator. Do not assume a flight-controller peripheral rail can supply the recording payload.

Measure loaded current, supply stability, startup behavior and usable battery energy before accepting endurance.

Reference: hardware_planning/flight-platform-comparison.md • The H7 and PM02 regulated outputs are not paralleled in that proposal.

PaveWatch | Acquisition and common reference

Record original measurements and clock information; associate accepted observations with aircraft pose.



Targets: ≥10 min recording; identify gaps >1 s; residual timing offset ≤20 ms. These are proposed acceptance limits.

No hardware-trigger or PTP capability is assumed in this diagram • Reference: final requirements SA1–SA3 / S6 and newer hardware notes.

Component comparison | Flight platform

Three purchase configurations across two airframes; aircraft budget excludes the LiDAR and recording payload.

Metric	X500 ARF + Kakute H7 Mini v1.5	S500 ARF + Kakute H7 Mini v1.5	S500 development kit / Pixhawk 6C
Aircraft parts subtotal	$657.17 • $42.83 headroom under $700	$587.17 • $112.83 headroom under $700	$669.20 • $30.80 headroom under $700
Why consider it	Best documented external payload rails / mounting provisions	Largest cost margin; extra allowance for charger and integration	Matched flight electronics; lower custom wiring effort
Integration work	Custom controller plate, harnesses, current/RC/pose links	Custom controller plate plus payload mounting	Check installed payload fit; configure existing kit links
Common acceptance gate	≥10 min loaded acquisition flight + ≥20% usable energy reserve	Same loaded-flight gate; published unloaded hover is separate	Same loaded-flight gate; measure actual mass / current
Draft disposition	Planning document’s first choice for mounting; conditional on quote / fit	Budget-first alternative; strongest charger/fee margin	Simpler-integration alternative; tight delivered-cost margin

All three include a new 4S pack, matched RC, M10 navigation, telemetry and monitoring. Final choice awaits fit and endurance.

Source: hardware_planning/flight-platform-comparison.md, checked 23 Oct 2026 • Fees/charger and payload budget remain separate.

Component comparison | Recording computer

Use the revised compute candidates; benchmark concurrent LiDAR, camera and SSD capture on the complete stack.

Metric	Orange Pi Zero 3 + USB expansion	Raspberry Pi 4, 2 GB	Raspberry Pi Zero 2 W + Ethernet hub
Input / storage paths	Native Ethernet; USB camera via expansion; SSD on USB 2.0	Native Ethernet; USB 3.0 SSD; USB or CSI camera	Ethernet adapter and SSD share USB host link; CSI camera option
Board + I/O allowance	$40 board + $10 expansion = ~$50	$60 board; native Ethernet / USB ports	$15 board + $20 hub = ~$35
Main tradeoff	Compact cost-oriented recorder; validate OS, USB expansion and sustained writes	More storage/I/O flexibility; validate full capture mode and power	Shared-bus constraint and adapter complexity; availability unconfirmed
Required evidence	≥10 min simultaneous recording; gap detection; timing and power checks	Same recording/timing gate; USB 3.0 alone does not guarantee no gaps	Same gate under concurrent traffic; measure effective throughput
Diagram disposition	Candidate; model remains generic in system architecture	Candidate; model remains generic in system architecture	Candidate/fallback; no ordering choice implied

The source summary totals conflict with its itemized costs. No ~8–15 MB/s pipeline rate or loss-free capture is assumed.

Source: hardware_planning/compute.md • Itemized planning estimates used; camera, SSD, cooling and regulator excluded from this cost row.

Component comparison | Surface imagery

Use the revised camera shortlist; optical detail, exposure and interface mode must be evaluated at the survey envelope.

Metric	InnoMaker U20CAM-9281M	InnoMaker 1080p USB board	Arducam AR0234 USB family
Sensor / image type	OV9281 monochrome global shutter; up to 1280 × 800	Color 1080p candidate; rolling-shutter description in notes	Color global-shutter candidate; exact USB variant / lens pending
Planning price band	~$30–35 in planning notes	~$15–20 in planning notes; exact SKU needed	~$80–95 in planning notes; exact SKU needed
Why consider it	Global capture timing and monochrome output; lens / detail tradeoff	Lower camera allowance and color context; exposure/readout characterization	Higher-resolution global-shutter option; USB mode / cost tradeoff
Validation needed	Exposure, focus, FOV, ground detail and timing offset	Same tests, plus rolling-shutter distortion at actual motion	Same optical tests; verify supported full-resolution USB mode
Diagram disposition	Candidate; sensor is 1 MP, not limited to the notes' "720p" label	Candidate; model remains generic until vendor SKU is chosen	Candidate; select exact product before assuming Orange Pi compatibility

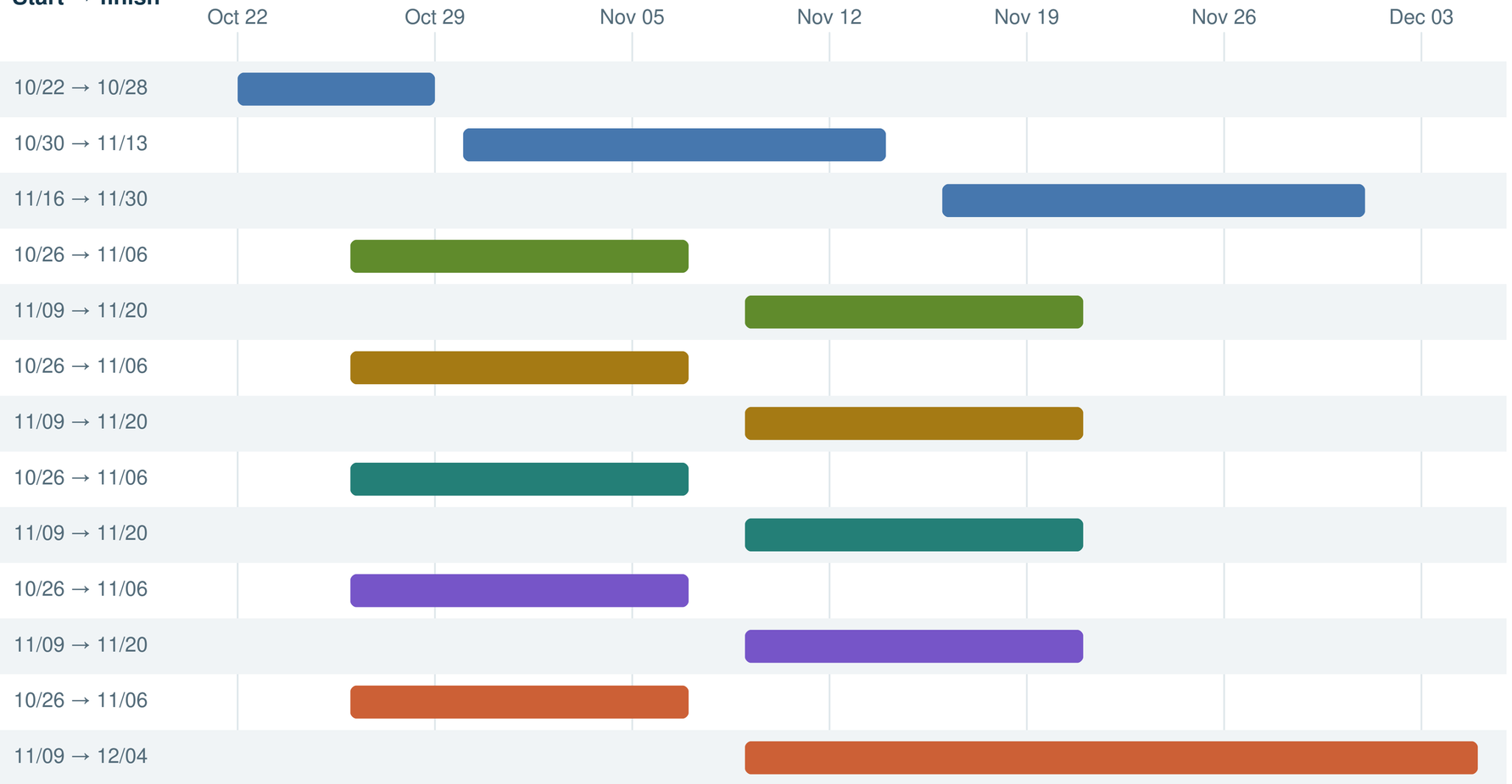
Global shutter removes rolling-shutter skew; motion blur still depends on exposure. Model and lens remain undecided.

Sources: hardware_planning/camera.md; InnoMaker U20CAM-9281M manufacturer repository; Arducam B0495 reference datasheet.

Capstone I | Subsystem plan

Proposed Fall work from the current draft; hardware and field milestones depend on delivery and approval.

Subsystem / owner and activity	Start → finish
FP / Ankit — Freeze platform, 4S pack and mass budget	10/22 → 10/28
FP / Ankit — Procure, assemble and bench-check aircraft	10/30 → 11/13
FP / Ankit — Takeover / pose; conditional loaded flights	11/16 → 11/30
DT / Shisir — Health schema and replay	10/26 → 11/06
DT / Shisir — Status latency and injected-fault tests	11/09 → 11/20
SA / Abhinav — Receive sensor; capture / decode packets	10/26 → 11/06
SA / Abhinav — Recorder + independent timing / range checks	11/09 → 11/20
GC / Pritam — Site request; navigation / links / route plan	10/26 → 11/06
GC / Pritam — Waypoints, acknowledgements and buffer tests	11/09 → 11/20
PC / Aalok — Common frame, fixtures and export path	10/26 → 11/06
PC / Aalok — Held-out analysis; three ≥1 GB exports	11/09 → 11/20
RD / Saugat — Finding schema and evidence interface	10/26 → 11/06
RD / Saugat — Warnings / evidence integration and release	11/09 → 12/04



Milestones: Oct 23 MSDR • Oct 30 approach draft • Nov 13 approach final • Nov 20 rubric • Nov 30 demo • Dec 4 final document

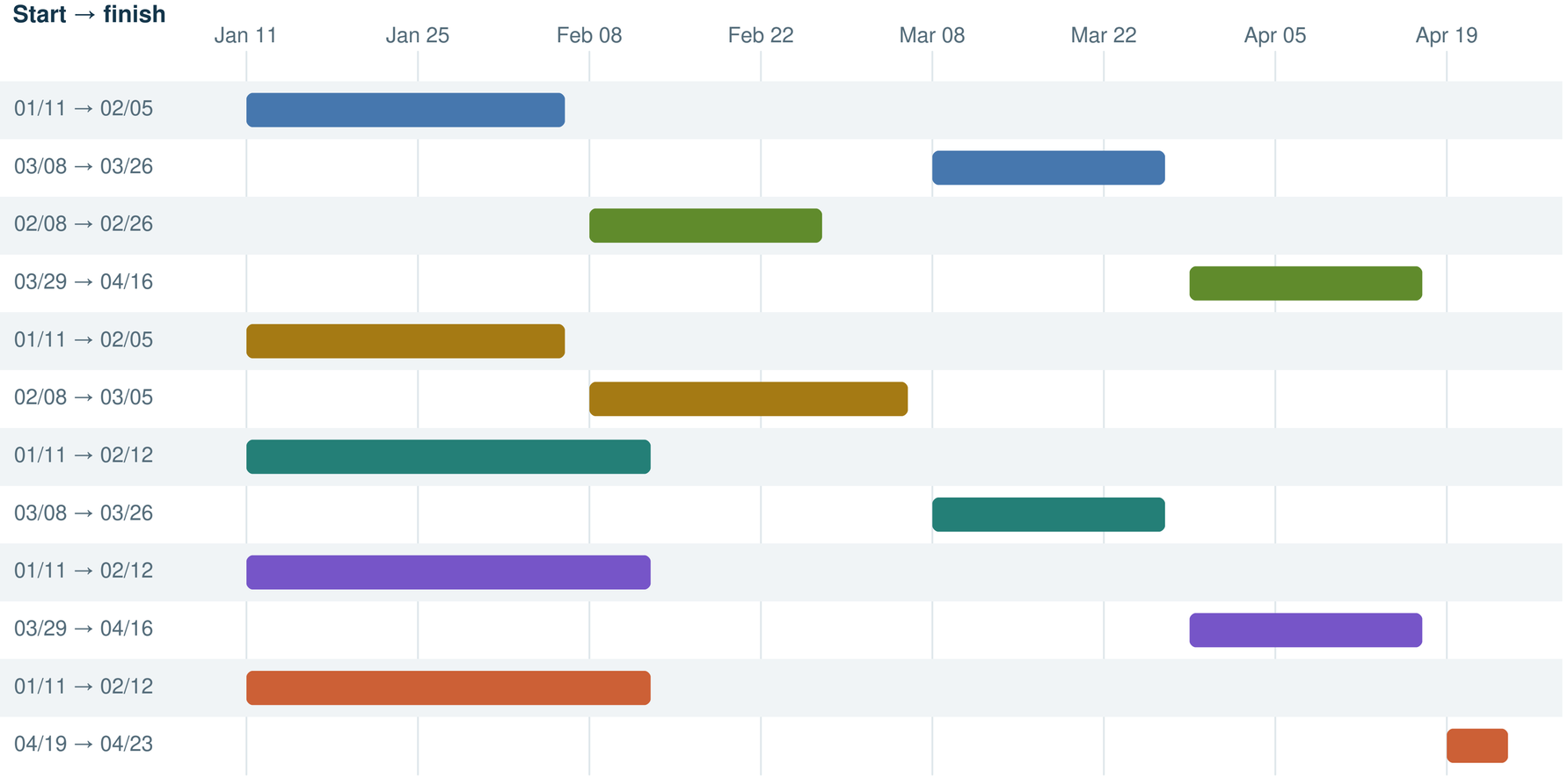
Each bar is a proposed task window, not a completion claim. Dates and dependencies need owner review.

Source: MSDR/PaveWatch_MSDR_Schedule.csv • Diagram text follows newer platform research; legacy RTK wording is treated as optional.

Capstone II | Subsystem plan

January–April 2027 dates are planning assumptions; confirm against the Capstone II calendar.

Subsystem / owner and activity	Start → finish
FP / Ankit — Finish loaded platform integration	01/11 → 02/05
FP / Ankit — Endurance, route and repeat survey flights	03/08 → 03/26
DT / Shisir — Integrated health and fault characterization	02/08 → 02/26
DT / Shisir — Field-log audit and acceptance evidence	03/29 → 04/16
SA / Abhinav — Camera, mounts and timing calibration	01/11 → 02/05
SA / Abhinav — Controlled-motion and envelope checks	02/08 → 03/05
GC / Pritam — Navigation, mission and link integration	01/11 → 02/12
GC / Pritam — Field missions and recovery evidence	03/08 → 03/26
PC / Aalok — Geometry / image pipeline and held-out protocol	01/11 → 02/12
PC / Aalok — Field errors, fusion and optional repeat survey	03/29 → 04/16
RD / Saugat — Reviewer interface and geographic exports	01/11 → 02/12
RD / Saugat — Acceptance package and reproducible release	04/19 → 04/23



Sequence: integrate and calibrate → controlled motion → authorized loaded flights → held-out evaluation → final evidence release

Each bar is a proposed task window, not a completion claim. Dates and dependencies need owner review.

Source: MSDR/PaveWatch_MSDR_Schedule.csv • Diagram text follows newer platform research; legacy RTK wording is treated as optional.

Capstone I deliverables | Acquisition and analysis

Planned Capstone I evidence, owners, measurable targets and the artifact that will be shown.

Subsystem / owner	Acceptance target	Demonstration method and artifact
Abhinav Dhakal **Sensors / acquisition**	≥10 min range + images; flag gaps >1 s. Residual motion-reference timing ≤20 ms. ≥90% range errors ≤60 mm.	Bench capture + replay. Range targets at 5, 6.5 and 8 m; ≥10 observations per condition. Independent timing events at start/middle/end. Deliver recording, calibration and error plots.
Shisir Baral **Diagnostics / telemetry**	≥90% battery / mode / acquisition / storage updates reach operator ≤4 s over ≥10 min. Explicit sensor / pose-loss indications.	Timestamp ≥300 updates with a common clock; inject five >1 s gaps per stream and remove pose inputs. Deliver live/replay health view, latency distribution and gap records.
Aalok Uprety **Processing / export**	Frame checks plus S2/S3/S5 outcome metrics. Three ≥1 GB survey exports to laptop, each ≤30 min and integrity checked.	≥10 independent frame-check points; held-out geometry and image cases; known slopes. Score detection and measurement separately. Deliver analysis tables, source evidence and exported survey package.

Hardware-dependent limits require real measurements. Replay supports the software behavior actually exercised.

Source: final requirements including October 19 edits • Physical, simulated and replay evidence must retain their actual scope.

Capstone I deliverables | Control and reporting

Planned Capstone I evidence, owners, measurable targets and the artifact that will be shown.

Subsystem / owner	Acceptance target	Demonstration method and artifact
Ankit Dhakal **Flight / power / control**	Takeover ≤1 s, 10/10 trials. Timestamped pose ≥10 min. Loaded flight/route targets remain physical acceptance gates.	Simulator or propeller-free bench: commands, mode acknowledgements and known pose references. Physical evidence: three loaded ≥10 min flights with ≥20% reserve; five independently referenced passes.
Pritam Adhikari **Ground control / links**	≥10 waypoints entered/transferred/verified ≤10 min over five trials. ≥90% command acknowledgements ≤2 s. Define the 60 s buffer stream.	Simulator/bench route readback and command logs; ≥100 commands across stated 10/50/100 m conditions where authorized. Disconnect for 30–60 s; inspect bounded-stream recovery and ordering.
Saugat Karki **Reporting / evidence**	Show survey findings with associated images, available measurements, units, gap warnings and unusable-reference warnings.	Live geometry and image-only findings; compare displayed content with source results. Deliver reviewer interface, example report/export and a source-to-display checklist.

Review decisions: FP4 says 5° in its requirement and 1° in its test; GC3 stream scope and gradient-error limit remain open.

Source: final requirements including October 19 edits • Physical, simulated and replay evidence must retain their actual scope.